

EXTENSION 1

Stage 6

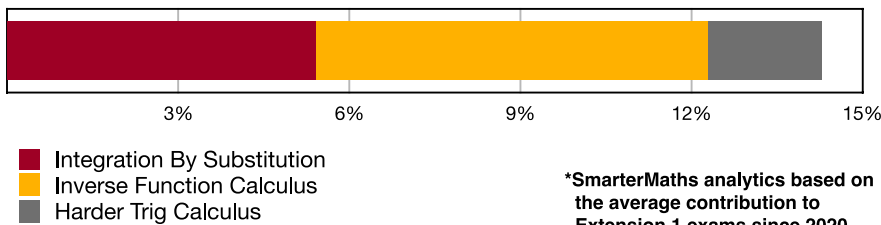
Calculus (Ext1), C2 Further Calculus Skills (Y12)

Harder Trig Calculus (Ext1)

Teacher: SHS Maths

Exam Equivalent Time: 45 minutes (based on allocation of 1.5 minutes per mark)

C2 Further Calculus Skills



HISTORICAL CONTRIBUTION

- C2 Further Calculus Skills* has contributed a substantial average of 14.3% per Ext1 new syllabus exam since its introduction in 2020.
- This topic has been split into three sub-categories for analysis purposes which are: *1-Integration By Substitution* (5.4%), *2-Inverse Function Calculus* (6.9%) and *3-Harder Trig Calculus* (2.0%).
- This analysis looks at *Harder Trig Calculus*.

HSC ANALYSIS - What to expect and common pitfalls

- Harder Trig Calculus* (2.0%) was most recently tested in 2024 and prior to that in both new syllabus Ext1 exams between 2020–21.
- The difficulty level of past questions (from the new and old syllabus) has ranged from a lower band 3–4 level, to very challenging band 6 applied questions involving tangent concepts.
- 2020 Ext1 13c saw *Inverse Trig Function Calculus* overlap with this topic area in a question that caused all sorts of problems and deserves specific attention.
- A key skill - students must be able to *competently and quickly* use the identities $\cos 2\theta = 1 - 2\sin^2\theta = 2\cos^2\theta - 1$. These are regularly examined and prone to silly errors - review carefully!
- Although not included in this analysis, this topic area provides core integration skills for later calculus topics such as Solids of Revolution.

Questions

1. Calculus, EXT1 C2 2021 HSC 2 MC

Which of the following integrals is equivalent to $\int \sin^2 3x \, dx$?

- A. $\int \frac{1 + \cos 6x}{2} dx$
- B. $\int \frac{1 - \cos 6x}{2} dx$
- C. $\int \frac{1 + \sin 6x}{2} dx$
- D. $\int \frac{1 - \sin 6x}{2} dx$

2. Calculus, EXT1 C2 2016 HSC 5 MC

Which expression is equal to $\int \sin^2 2x \, dx$?

- A. $\frac{1}{2} \left(x - \frac{1}{4} \sin 4x \right) + c$
- B. $\frac{1}{2} \left(x + \frac{1}{4} \sin 4x \right) + c$
- C. $\frac{\sin^3 2x}{6} + c$
- D. $\frac{-\cos^3 2x}{6} + c$

3. Calculus, EXT1 C2 2012 HSC 7 MC

Which expression is equal to $\int \sin^2 3x \, dx$?

- A. $\frac{1}{2} \left(x - \frac{1}{3} \sin 3x \right) + C$
- B. $\frac{1}{2} \left(x + \frac{1}{3} \sin 3x \right) + C$
- C. $\frac{1}{2} \left(x - \frac{1}{6} \sin 6x \right) + C$
- D. $\frac{1}{2} \left(x + \frac{1}{6} \sin 6x \right) + C$

4. Calculus, EXT1 C2 2019 HSC 11e

Find $\int 2\sin^2 4x \, dx$. (2 marks)

5. Calculus, EXT1 C2 2005 HSC 3b

i. By expanding the left-hand side, show that

$\sin(5x + 4x) + \sin(5x - 4x) = 2\sin(5x)\cos(4x)$ (1 mark)

ii. Hence find $\int \sin(5x)\cos(4x) \, dx$. (2 marks)

6. Calculus, EXT1 C2 2020 HSC 12d

Find $\int_0^{\frac{\pi}{2}} \cos 5x \sin 3x \, dx$. (3 marks)

7. Calculus, EXT1 C2 2024 HSC 13b

i. Show that $\cos^4 x + \sin^4 x = \frac{1 + \cos^2 2x}{2}$. (2 marks)

ii. Hence, or otherwise, evaluate $\int_0^{\frac{\pi}{4}} (\cos^4 x + \sin^4 x) \, dx$. (3 marks)

8. Calculus, EXT1 C2 2010 HSC 2a

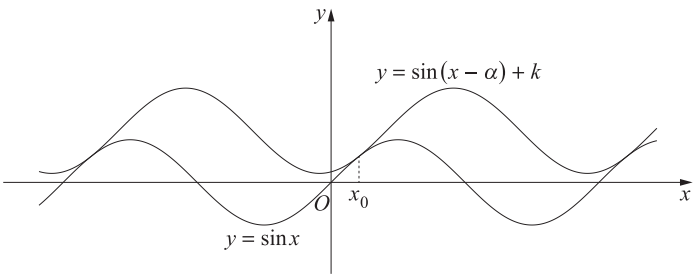
The derivative of a function $f(x)$ is given by

$f'(x) = \sin^2 x$.

Find $f(x)$, given that $f(0) = 2$. (2 marks)

9. Calculus, EXT1 C2 2019 HSC 14c

The diagram shows the two curves $y = \sin x$ and $y = \sin(x - \alpha) + k$, where $0 < \alpha < \pi$ and $k > 0$. The two curves have a common tangent at x_0 where $0 < x_0 < \frac{\pi}{2}$.



i. Explain why $\cos x_0 = \cos(x_0 - \alpha)$. (1 mark)

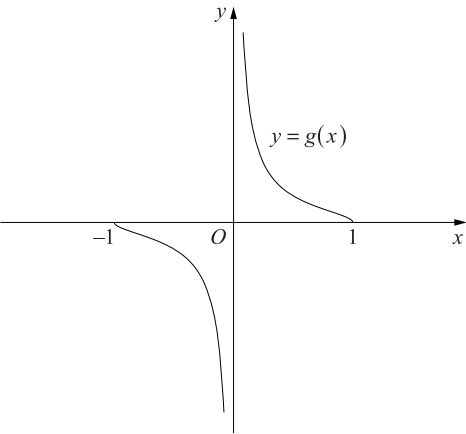
ii. Show that $\sin x_0 = -\sin(x_0 - \alpha)$. (2 marks)

iii. Hence, or otherwise, find k in terms of α . (2 marks)

10. Calculus, EXT1 C2 2020 HSC 13c

Suppose $f(x) = \tan(\cos^{-1}(x))$ and $g(x) = \frac{\sqrt{1-x^2}}{x}$.

The graph of $y = g(x)$ is given.



i. Show that $f'(x) = g'(x)$. (4 marks)

ii. Using part (i), or otherwise, show that $f(x) = g(x)$. (3 marks)

Worked Solutions

1. Calculus, EXT1 C2 2021 HSC 2 MC

$$\int \sin^2 3x \, dx = \frac{1}{2} \int (1 - \cos 6x) \, dx$$
$$\Rightarrow B$$

2. Calculus, EXT1 C2 2016 HSC 5 MC

$$\int \sin^2 2x \, dx$$
$$= \frac{1}{2} \int (1 - \cos 4x) \, dx$$
$$= \frac{1}{2} \left(x - \frac{1}{4} \sin 4x \right) + c$$
$$\Rightarrow A$$

3. Calculus, EXT1 C2 2012 HSC 7 MC

Using: $\sin^2 a = \frac{1}{2}(1 - \cos 2a)$

$$\int \sin^2 3x \, dx = \frac{1}{2} \int (1 - \cos 6x) \, dx$$
$$= \frac{1}{2} \left(x - \frac{1}{6} \sin 6x \right) + C$$
$$\Rightarrow C$$

4. Calculus, EXT1 C2 2019 HSC 11e

Using $\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta)$:

$$\int 2\sin^2 4x \, dx = \int 2 \times \frac{1}{2}(1 - \cos 8x) \, dx$$
$$= \int 1 - \cos 8x \, dx$$
$$= x - \frac{1}{8} \sin 8x + C$$

Worked Solutions

5. Calculus, EXT1 C2 2005 HSC 3b

i. $\sin(5x + 4x) + \sin(5x - 4x) = 2\sin(5x)\cos(4x)$

$$\text{LHS} = \sin(5x)\cos(4x) - \sin(4x)\cos(5x) + \sin(5x)\cos(4x) + \sin(4x)\cos(5x)$$
$$= 2\sin(5x)\cos(4x) \quad \dots \text{ as required}$$

ii. $\int \sin(5x)\cos(4x) \, dx$

$$= \frac{1}{2} \int 2\sin(5x)\cos(4x) \, dx$$
$$= \frac{1}{2} \int \sin(5x + 4x) + \sin(5x - 4x) \, dx$$
$$= \frac{1}{2} \int \sin(9x) + \sin(x) \, dx$$
$$= \frac{1}{2} \left[-\frac{1}{9} \cos(9x) - \cos(x) \right] + c$$
$$= -\frac{1}{18} \cos(9x) - \frac{1}{2} \cos(x) + c$$

6. Calculus, EXT1 C2 2020 HSC 12d

$$\int_0^{\frac{\pi}{2}} \cos 5x \sin 3x \, dx = \frac{1}{2} \int_0^{\frac{\pi}{2}} 2\cos 5x \sin 3x \, dx$$
$$= \frac{1}{2} \int_0^{\frac{\pi}{2}} \sin 8x - \sin 2x \, dx$$
$$= \frac{1}{2} \left[-\frac{1}{8} \cos 8x + \frac{1}{2} \cos 2x \right]_0^{\frac{\pi}{2}}$$
$$= \frac{1}{2} \left[\left(-\frac{1}{8} \cos 4\pi + \frac{1}{2} \cos \pi \right) - \left(-\frac{1}{8} \cos 0 + \frac{1}{2} \cos 0 \right) \right]$$
$$= \frac{1}{2} \left(-\frac{1}{8} - \frac{1}{2} + \frac{1}{8} - \frac{1}{2} \right)$$
$$= -\frac{1}{2}$$

7. Calculus, EXT1 C2 2024 HSC 13b

i.
$$\begin{aligned} \text{LHS} &= \left[\frac{1}{2} (1 + \cos(2x)) \right]^2 + \left[\frac{1}{2} (1 - \cos(2x)) \right]^2 \\ &= \frac{1}{4} (1 + 2\cos(2x) + \cos^2(2x) + 1 - 2\cos(2x) + \cos^2(2x)) \\ &= \frac{1}{4} (2 + 2\cos^2(2x)) \\ &= \frac{1 + \cos^2(2x)}{2} \end{aligned}$$

ii.
$$\begin{aligned} &\int_0^{\frac{\pi}{4}} (\cos^4 x + \sin^4 x) dx \\ &= \frac{1}{2} \int_0^{\frac{\pi}{4}} 1 + \cos^2(2x) dx \\ &= \frac{1}{2} \int_0^{\frac{\pi}{4}} 1 + \frac{1}{2} (1 + \cos(4x)) dx \\ &= \frac{1}{2} \left[\frac{3}{2} x + \frac{1}{8} \sin(4x) \right]_0^{\frac{\pi}{4}} \\ &= \frac{1}{2} \left[\frac{3}{2} \times \frac{\pi}{4} + \frac{1}{8} \sin \pi - 0 \right] \\ &= \frac{3\pi}{16} \end{aligned}$$

8. Calculus, EXT1 C2 2010 HSC 2a

$$\begin{aligned} f'(x) &= \sin^2 x \\ f(x) &= \int \sin^2 x \, dx \\ &= \int \frac{1}{2} (1 - \cos 2x) \, dx \\ &= \int \frac{1}{2} - \frac{1}{2} \cos 2x \, dx \\ &= \frac{1}{2} x - \frac{1}{4} \sin 2x + c \end{aligned}$$

Given $f(0) = 2$,

$$\begin{aligned} 2 &= \frac{1}{2} \times 0 - \frac{1}{4} \sin 0 + c \\ \therefore c &= 2 \end{aligned}$$

$$\therefore f(x) = \frac{1}{2} x - \frac{1}{4} \sin 2x + 2$$

9. Calculus, EXT1 C2 2019 HSC 14c

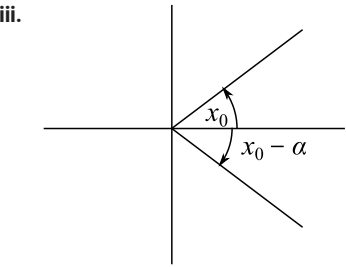
i.
$$\begin{aligned} y_1 &= \sin x \\ \frac{dy_1}{dx} &= \cos x \\ y_2 &= \sin(x - \alpha) + k \\ \frac{dy_2}{dx} &= \cos(x - \alpha) \end{aligned}$$

At $x = x_0$, tangent is common
 $\therefore \cos x_0 = \cos(x_0 - \alpha)$

◆ Mean mark part (i) 47%.

ii. x_0 is in 1st quadrant (given)
Using part (i):
 $\cos x_0 = \cos(x_0 - \alpha) > 0$
 $\Rightarrow x_0 - \alpha$ is in 4th quadrant ($0 < \alpha < \pi$)
Since sin is positive in 1st quadrant and negative in 4th quadrant
 $\Rightarrow \sin x_0 = -\sin(x_0 - \alpha)$

◆◆ Mean mark part (ii) 19%.



When $x = x_0$,

$$\begin{aligned} y_1 &= \sin x_0 \\ y_2 &= \sin(x_0 - \alpha) + k \\ \sin x_0 &= \sin(x_0 - \alpha) + k \\ &= -\sin x_0 + k \\ k &= 2 \sin x_0 \end{aligned}$$

Since $\cos x_0 = \cos(x_0 - \alpha)$

$$\begin{aligned} x_0 &= -(x_0 - \alpha) \\ 2x_0 &= \alpha \\ x_0 &= \frac{\alpha}{2} \end{aligned}$$

◆◆ Mean mark part (iii) 21%.

$$\therefore k = 2\sin \frac{\alpha}{2}$$

10. Calculus, EXT1 C2 2020 HSC 13c

i. $f(x) = \tan(\cos^{-1}(x))$

◆ Mean mark (i) 50%.

$$\begin{aligned} f'(x) &= -\frac{1}{\sqrt{1-x^2}} \cdot \sec^2(\cos^{-1}(x)) \\ &= -\frac{1}{\sqrt{1-x^2}} \cdot \frac{1}{\cos^2(\cos^{-1}(x))} \\ &= -\frac{1}{x^2\sqrt{1-x^2}} \end{aligned}$$

$$g(x) = (1-x^2)^{\frac{1}{2}} \cdot x^{-1}$$

$$\begin{aligned} g'(x) &= \frac{1}{2} \cdot -2x(1-x^2)^{-\frac{1}{2}} \cdot x^{-1} - (1-x^2)^{\frac{1}{2}} \cdot x^{-2} \\ &= \frac{-x}{x\sqrt{1-x^2}} - \frac{\sqrt{1-x^2}}{x^2} \\ &= \frac{-x^2 - \sqrt{1-x^2}\sqrt{1-x^2}}{x^2\sqrt{1-x^2}} \\ &= \frac{-x^2 - (1-x^2)}{x^2\sqrt{1-x^2}} \\ &= -\frac{1}{x^2\sqrt{1-x^2}} \\ &= f'(x) \end{aligned}$$

ii. $f'(x) = g'(x)$

$$\Rightarrow f(x) = g(x) + c$$

◆◆◆ Mean mark (ii) 15%.

Find c :

$$f(1) = \tan(\cos^{-1}1)$$

$$= \tan 0$$

$$= 0$$

$$g(1) = \frac{\sqrt{1-1}}{0} = 0$$

$$f(1) = g(1) + c$$

$$\therefore c = 0$$

$$\therefore f(x) = g(x)$$